# UNIX Lab Manual - Part B

## AWK Program 1

```
BEGIN{
FS=","
printf("
    -----------------------------------------------------------\n")
    ;
printf("%-4s\t%-15s\t%-8s\n","ID","Name","Revenue");
printf("
    -----------------------------------------------------------\n")
    ;
}$3>50 && $4>10000{
printf("%-4d\t%-15s\t%-.2f\n",$1,$2,$4);
tr=tr+$4;
}
END{
printf("
    -----------------------------------------------------------\n")
    ;
printf("Total Revenue generated by selected employees=%-.2f\n",tr)
    ;
printf("
    -----------------------------------------------------------\n")
    ;
}
```

## AWK Program 2

```
BEGIN {
print("Processing Numbers in a Text file....\n");
}
{
    split($0, a, " ");

    for (i = 1; i <= NF; i++) {
        row[NR] += a[i];
    }

    printf "%s ", $0;
    printf "%d\n", row[NR];
```

```
}
END {
print("Script ended...");
}
```

## AWK Program 3

```
BEGIN{
FS="|"
printf("----------------------------------------------------\n")
printf("Salary Details of Sales and Marketing\n")
printf("----------------------------------------------------\n")
}
$4=="sales"||$4=="marketing"{
da=0.25*$6
hra=0.5*$6
gp=$6+da+hra
total[1]=total[1]+$6
total[2]=total[2]+da
total[3]=total[3]+hra
total[4]=total[4]+gp
count++;
}
END{
printf("\t%-10s%-10s%-10s%-10s\n","Basic","DA","HRA","Gross")
printf("Total:\t%-10d%-10d%-10d%-10d\n",total[1],total[2],total
    [3],total[4])
printf("Avg:\t%-10.2f%-10.2f%-10.2f%-10.2f\n",total[1]/count,total
    [2]/count,total[3]/count,total[4]/count)
printf("----------------------------------------------------\n")
}
```

## Perl Program 1

```
#!/usr/bin/perl
print "Palindrome check through command line arguments\n";

if ($#ARGV < 0) {
    print "Enter valid numbers as command line arguments\n";
    exit;
}

foreach $num (@ARGV) {
    if ($num !~ /^\d+$/) {
        print "Invalid number: $num\n";
        next;
    }
```

```
    $sum = 0;
    $a = $num;

    until ($num == 0) {
        $y = $num % 10;
        $sum = $sum * 10 + $y;
        $num = int($num / 10);
    }

    if ($a == $sum) {
        print "$a is a palindrome number\n";
    } else {
        print "$a is not a palindrome number\n";
    }
}
```

## Perl Program 2

```
#!/usr/bin/perl
print "Sum of digits through command line arguments\n";

if ($#ARGV < 0) {
    print "Enter valid numbers as command line arguments\n";
    exit;
}

foreach $num (@ARGV) {
    if ($num !~ /^\d+$/) {
        print "Invalid number: $num\n";
        next;
    }

    $sum = 0;
    $a = $num;

    until ($num == 0) {
        $y = $num % 10;
        $sum = $sum + $y;
        $num = int($num / 10);
    }

    print "Sum of digits in $a is: $sum\n";
}
```